Tutorial: Software Installation

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- DBMS: [PostgreSQL](#)
- Client: [DataGrip](#)

Part 1. Environment Configuration

macOS users

1. Installation & Usage

Here are alternative ways to install PostgreSQL.

By Homebrew (Recommended)

Step 1. Install [Homebrew](#).

(If you already have it installed, skip this step.)

1. Prerequisites according to [Requirements](#):
   1. A 64-bit Intel CPU or Apple Silicon CPU [1](#)
   2. macOS Mojave (10.14) (or higher) [2](#)
   3. Command Line Tools (CLT) for Xcode: `xcode-select --install`, [developer.apple.com/downloads](#) or [Xcode](#) [3](#)
   4. A Bourne-compatible shell for installation (e.g. `bash` or `zsh`) [4](#)
2. Open the “Terminal” application
3. Enter the following command into a single line of the terminal

   ```
   /usr/bin/ruby -e "$(curl -fsSL
   https://raw.githubusercontent.com/Homebrew/install/master/install)"
   ```

4. If you meet the problem: "Failed to connect to raw.githubusercontent.com port 443: Operation". You can solve it by the [link](#).

如果有用mac的同学在装完homebrew后，安装postgresql很慢，可以尝试更换homebrew的源，例如：https://mirrors.tuna.tsinghua.edu.cn/help/homebrew/

Step 2. Install PostgreSQL.

1. Enter the command to update brew home.

```
brew update
```

2. Enter the command to install PostgreSQL.

```
brew install postgresql
```

3. Check your PostgreSQL version. (Check successfully installed)

```
postgres --version
```

4. Note that, after the initial installation, it would generate two elements:

- a database named `postgres`.
- a database user named *your system user name*.

Step 3. Run `brew info postgres` for details.

1. Manually (will not start after system startup), in command line:

```
pg_ctl -D /usr/local/var/postgres start # To start
pg_ctl -D /usr/local/var/postgres stop  # To stop
```

2. Automatically (will start after system startup), in command line:

```
brew services start postgresql  # To start
brew services stop postgresql   # To stop
```

By Postgres.app

It is a brand new installation method. It's really simple but there may be some potential issues. Visit PostgresApp for details and optional versions.

Step 1. Installation

1. Download the .dmg file. (Be patient...)
2. Mount the file, move Postgres to Applications folder, and double click Postgres in Applications folder. (Now the dmg file is useless and can be removed.)
3. Click "Initialize" to create a new server.
4. You can change path and port in "Server settings" when stopped.

Step 2. Start & Stop

1. Click "Start" to start the server.
2. When server is started, you can open a command line client connected to a schema by double clicking this schema.
3. Click "Stop" to stop the server.

By Installer

1. Go to Postgresql Download Page, download installer. When the wizard prompts you to choose where to install PostgreSQL, point it to the **apps** subdirectory of your i.e. /Library/PostgreSQL/12.
2. Keep track of the **database superuser** name and **password**. You'll need these to initially create the LabKey database, the LabKey database user, and grant that user the owner role.
3. Keep track of the **database port**. (5432 for default)

By Docker

See For Docker users

2. Uninstallation

By Homebrew

```
brew uninstall postgres
```

By Postgres.app

1. Open *Finder*.
2. Go to *Applications*.
3. Move *Postgres.app* to *Trash*.

By Installer

There is a uninstall-postgresql.app in your installation directory. (i.e. /Library/PostgreSQL/12)

```
open /Library/PostgreSQL/12/uninstall-postgresql.app
```

For rest files, see this sof answer.

By Docker

See For Docker users

Linux users

1. Installation

By Package manager

Take **Ubuntu** as an example.

Step 1. Run the following command to access each URL in the source list, read the software list, and save it on the local computer.

```
sudo apt update
```

Step 2. Install PostgreSQL client first

```
sudo apt install postgresql-client
```

Step 3. Then Install PostgreSQL server

```
sudo apt install postgresql
```

Step 4. Check your PostgreSQL version. (Check successfully installed)

```
psql --version
```

Step 5. Enable and start `postgresql.service`

On Ubuntu, generally the PostgreSQL server will have an initialized database and automatically listen to port 5432 after the installation. (Other Linux distributions can behave differently and may require a manual database initialization.) As Ubuntu employs systemd to manage system services nowadays, you can check the status of the PostgreSQL service with `systemctl`:

```
systemctl status postgresql.service
```

As shown below, if `postgresql.service` is active, it is running in the background; If `postgresql.service` is enabled, it launches after system startup.

```
● postgresql.service - PostgreSQL RDBMS
     Loaded: loaded (/lib/systemd/system/postgresql.service; enabled; vendor preset: e>
     Active: active (exited) since Mon 2024-02-19 15:53:32 CST; 12min ago
    Process: 1993495 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 1993495 (code=exited, status=0/SUCCESS)
        CPU: 1ms

Feb 19 15:53:32 ecs-fa2a systemd[1]: Starting PostgreSQL RDBMS...
Feb 19 15:53:32 ecs-fa2a systemd[1]: Finished PostgreSQL RDBMS.
```

Enable `postgresql.service` if it is disabled:

```
sudo systemctl enable postgresql.service
```

Start `postgresql.service` if it is not running:

```
sudo systemctl start postgresql.service
```

Step 6. Note that, after the initial installation, it would generate three elements:

1. a database named `postgres`.
2. a database user named `postgres`.
3. a Linux system user named `postgres`.

Step 7. Change system user to postgres

```
su - postgres
```

By Docker

See For Docker users

2. Uninstallation

By Package manager

```
sudo apt remove postgresql postgresql-client
```

Note that your data files at `/var/lib/postgresql/` might be kept until you `purge` them.

By Docker

See For Docker users

3. Upgrade

Be careful when you want to upgrade your PostgreSQL server to a new **major-version**, especially if you are running a Linux distribution with a rolling release model (e.g., Arch Linux, Manjaro, openSUSE Tumbleweed). Your PostgreSQL server might fail to start after an upgrade because it might require data files in new storage format, which can be incompatible with your existing data in legacy format.

You can follow the instructions in PostgreSQL documentation to migrate existing data to the new format using tools like `pg_upgrade`.

Windows users

1. Installation

By Installer (Recommended)

1. Go to Postgresql Download Page, download installer. When the wizard prompts you to choose where to install PostgreSQL, point it to the **apps** subdirectory of your i.e. C:\labkey\apps\postgresql-10.6\
2. Keep track of the **PostgreSQL Windows Service** account name and password. LabKey Server needs to ask for it so that we can pass it along to the PostgreSQL installer.
3. Keep track of the **database superuser** name and password. You'll need these to initially create the LabKey database, the LabKey database user, and grant that user the owner role.

By Chocolatey

If you don't have chocolatey, go and get one.

```
choco install postgresql
```

By Docker

See For Docker users

2. Uninstallation

Universal way (by installer or choco)

1. Click "Start Menu", Go to "Settings" > "Apps" > "Apps & features".
2. Select "PostgreSQL", click "Remove".

By Chocolatey

```
choco uninstall postgresql
```

By Docker

See For Docker users

Docker users

Back to Environment Configuration

If you don't have Docker environment, please choose another installation method.

1. Installation

In command line:

```
docker run --name some-postgres -p 5432:5432 -e
POSTGRES_PASSWORD=mysecretpassword -d postgres
```

Note:

- You may need root privilege to run the above command if the current system user is not a member of the `docker` group.
- By default, Docker persists the data created by a postgres container [in an anonymous docker volume](#) on the host machine. You can bind a local directory to store the data by passing an `-v /path/to/dir:/var/lib/postgresql/data` argument.

2. Uninstallation

If your postgres container names "some-postgres"

```
docker stop some-postgres # Stop container
docker rm some-postgres   # Remove container
docker rmi postgres       # Remove image
```

Note that volumes are not deleted when containers are removed, unless the `--volumes` option is supplied to `docker rm`.

Part 2. How to use PostgreSQL

1. Prepare Parameters for Database Connection:

Generally, to visit database server, we needs following parameters:

- **Host IP** The IP address of server. `localhost` or `127.0.0.1` and represents the local ip of your computer.
- **Port**: The port of server. The default port of postgreSQL is `5432` .
- **User**: The user of database.
- **Password**: The password of database user.
- **Database**: The database you will to visit. The default database of postgreSQL always be `postgres`

Configuration on macOS

1. `⌘`+`space` to search and open "Terminal", then input the following command to access your database:

   ```
   psql postgres
   ```

   The system prompt would be postgres=#, which means you have connected to `postgres` database.

   (Note: `psql` is a command line PostgreSQL client program)

2. Find all roles in PostgreSQL

   ```
   \du
   ```

It will returns: (`yuemingzhu` is my database user)

```
postgres=# \du
                                   List of roles
 Role name  |                         Attributes                         |
Member of
------------+------------------------------------------------------------+-
----------
 yuemingzhu | Superuser, Create role, Create DB, Replication, Bypass RLS |
{}

(END)
```

3. Initial parameters on macOS:
   - **Host IP**: `localhost`
   - **Port**: `5432`
   - **User**: `yuemingzhu` (change it for your username)
   - **Password**: *null*
   - **Database**: `postgres`

Configuration on Linux

1. Connect to database `postgres` as `postgres` user:

   ```
   sudo -u postgres psql postgres
   ```

   The prompt would become `postgres=#`, which means you have connected to `postgres` database.

   (Note: `psql` is a command line PostgreSQL client program)

2. List all roles in PostgreSQL:

   ```
   postgres=# \du
                                      List of roles
    Role name |                         Attributes                         |
   Member of
   -----------+------------------------------------------------------------+--
   ---------
    postgres  | Superuser, Create role, Create DB, Replication, Bypass RLS |
   {}
   ```

3. Initial parameters on Linux:
   - **Host IP**: `localhost`
   - **Port**: `5432`
   - **User**: `postgres`
   - **Password**: *null*

- **Database**: `postgres`

Configuration on Windows

1. Initial parameters of Windows User:
    - **Host IP**: `localhost`
    - **Port**: `5432`
    - **User**: `postgres`
    - **Password**: *The password you set during the installation process*
    - **Database**: `postgres`

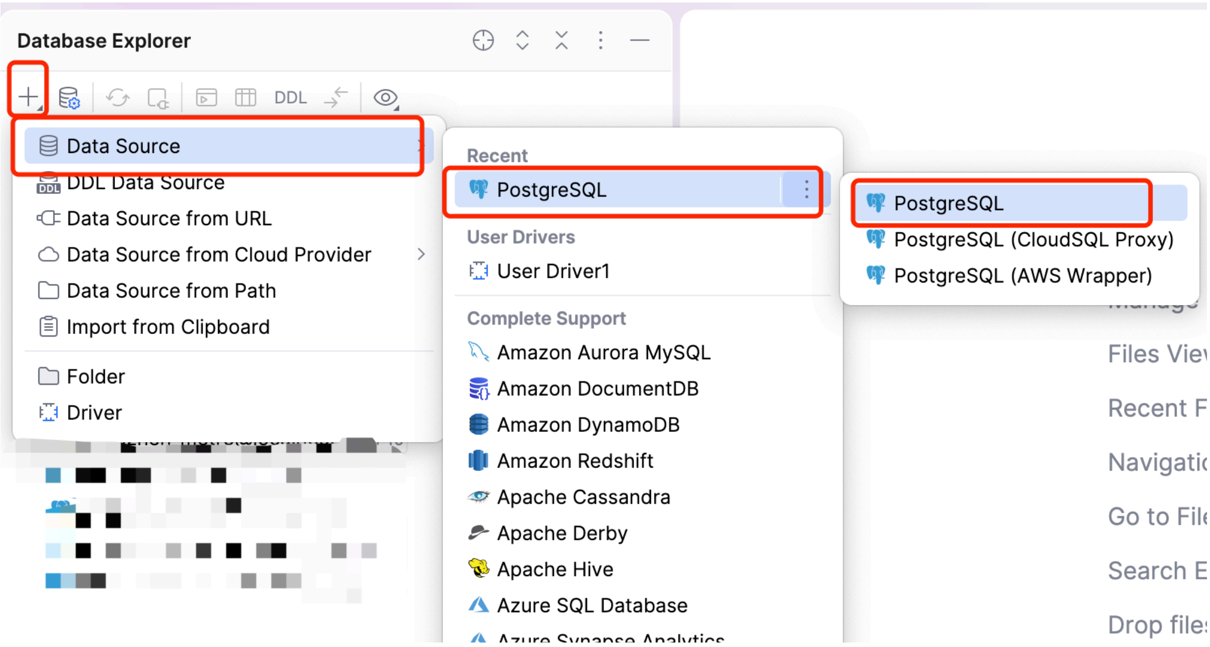
2. Datagrip

PostgreSQL is a server, we can visit the server by command, by a script or by a GUI program. In this case, DataGrip is a functional client with GUI platform, and in this course, we recommend you use Datagrip as client program.

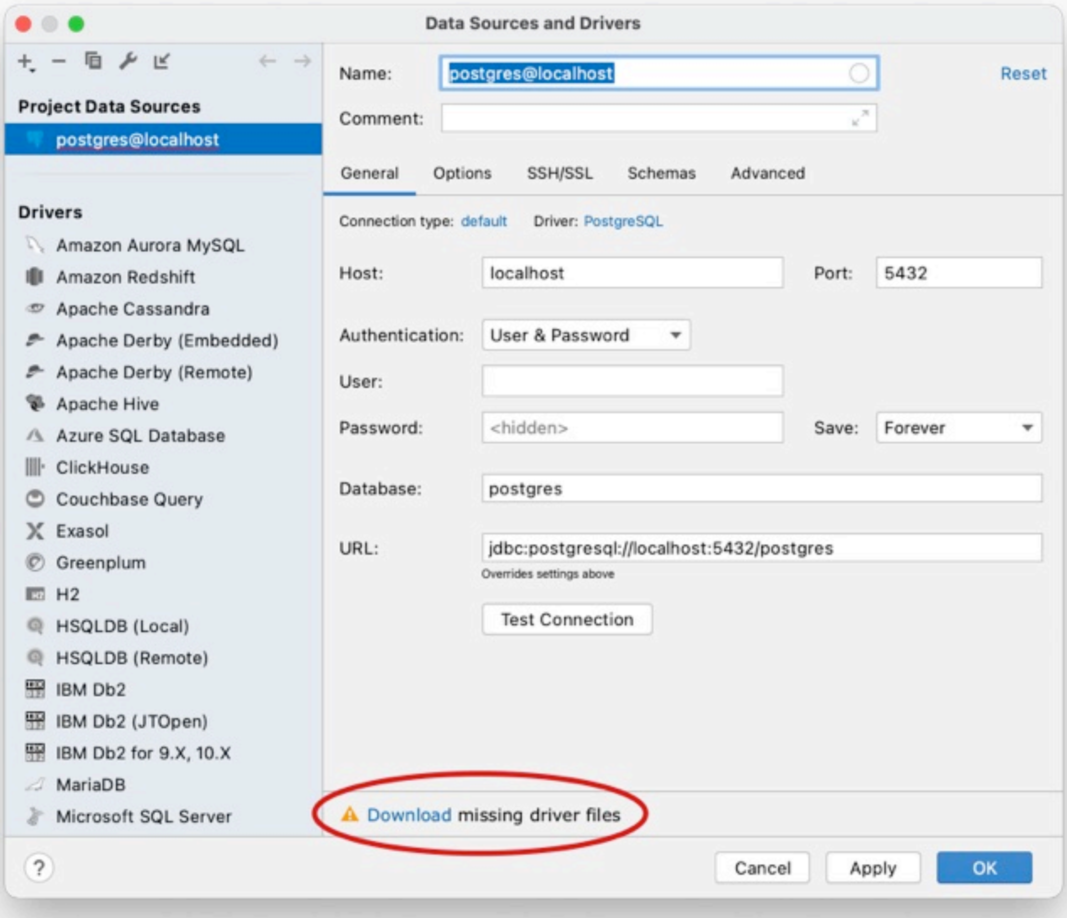
Here to [download DataGrip](), and then install it.

How to use DataGrip?

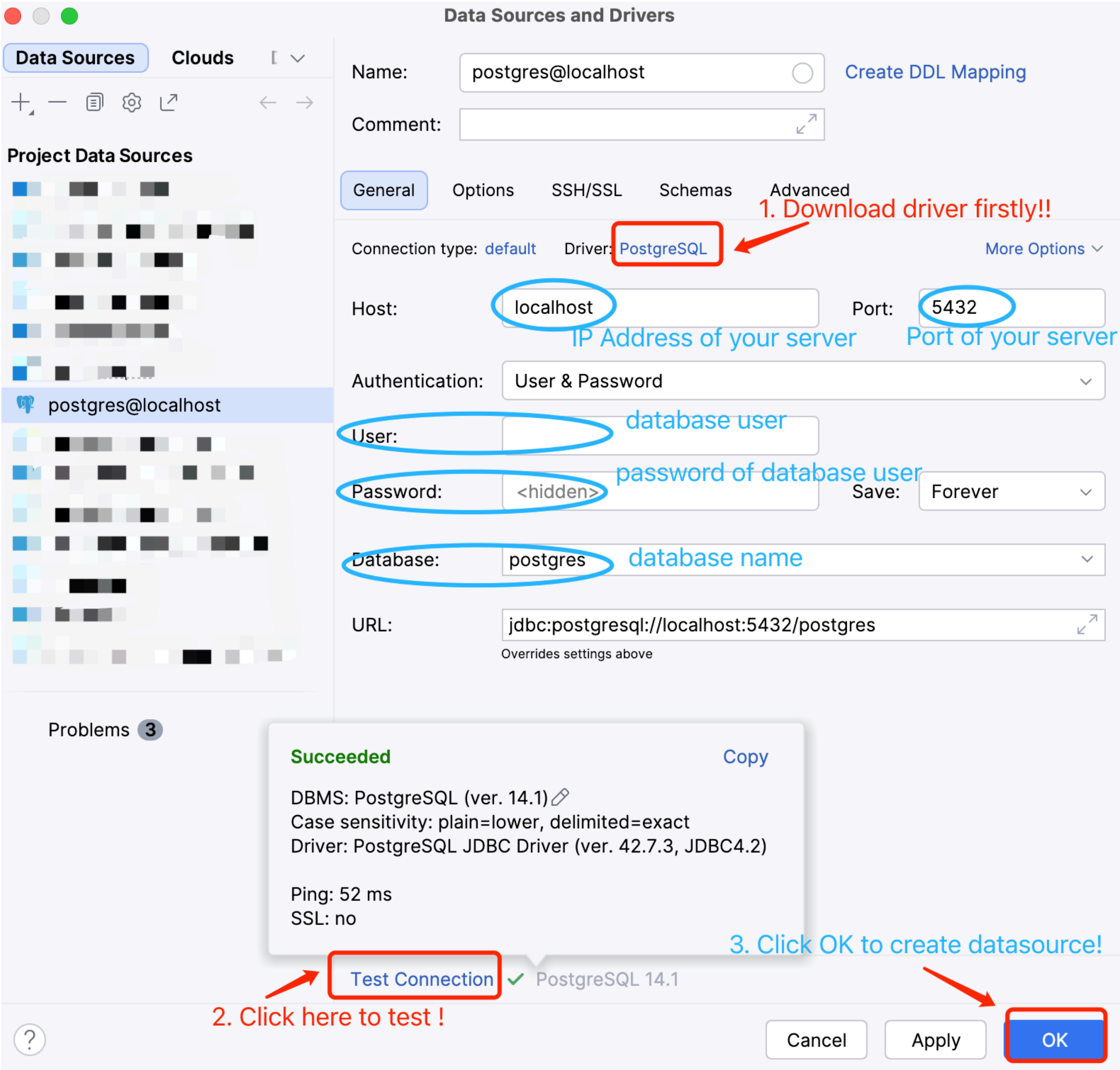
1. **Select Data Source**
2. **Add a client**



3. **Download the JDBC driver**

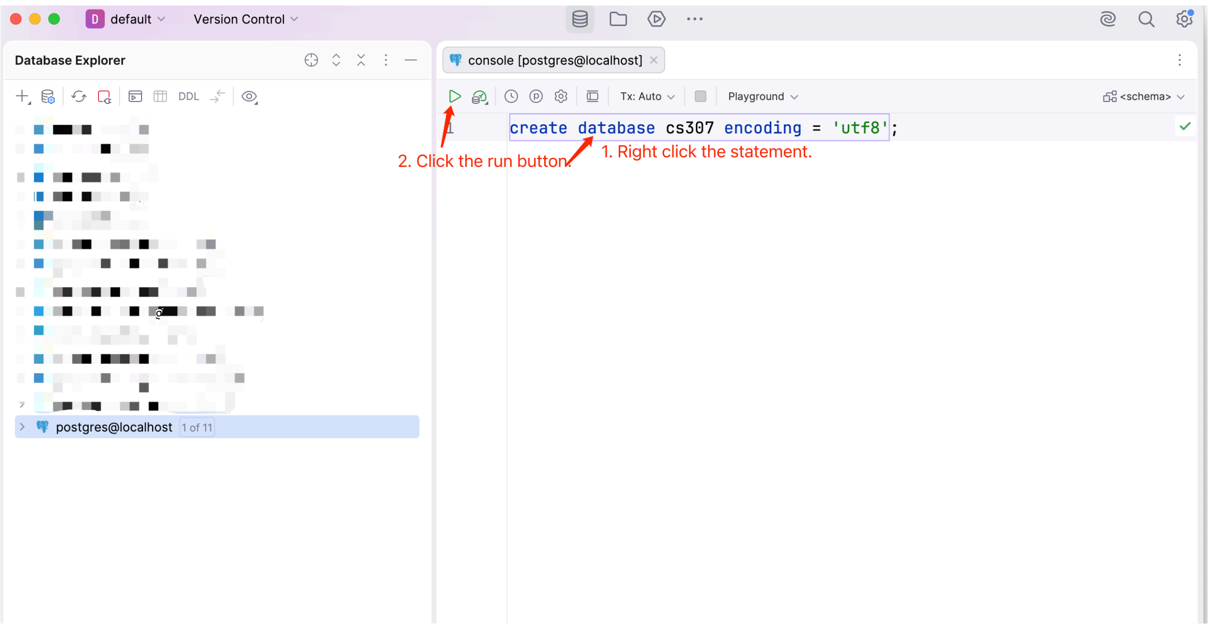


4. **Fill in Host, User, Password and Database, and then click Test Connection**



5. **Try to create a database in DataGrip**

```
create database cs307 encoding='utf8';
```



6. **Find all databases**

```sql
select datname from pg_database;
```

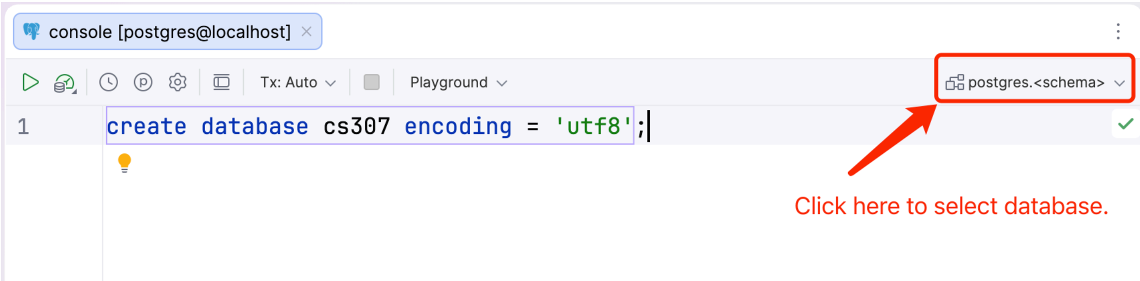
7. **Try to create a superuser**

```sql
create user checker superuser password '123456';
```

A new user named `checker` has been created with the password `123456`

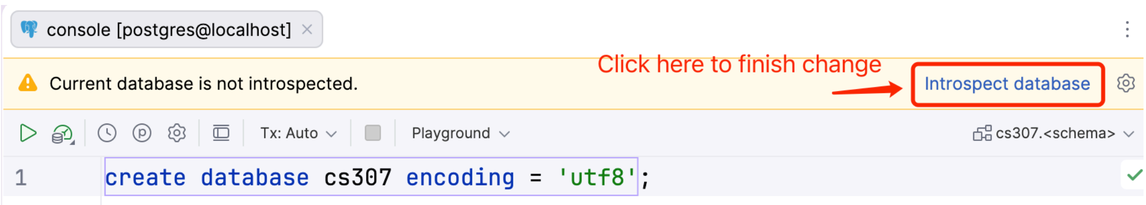
8. **How to check current schema and change database?**

   - Step1

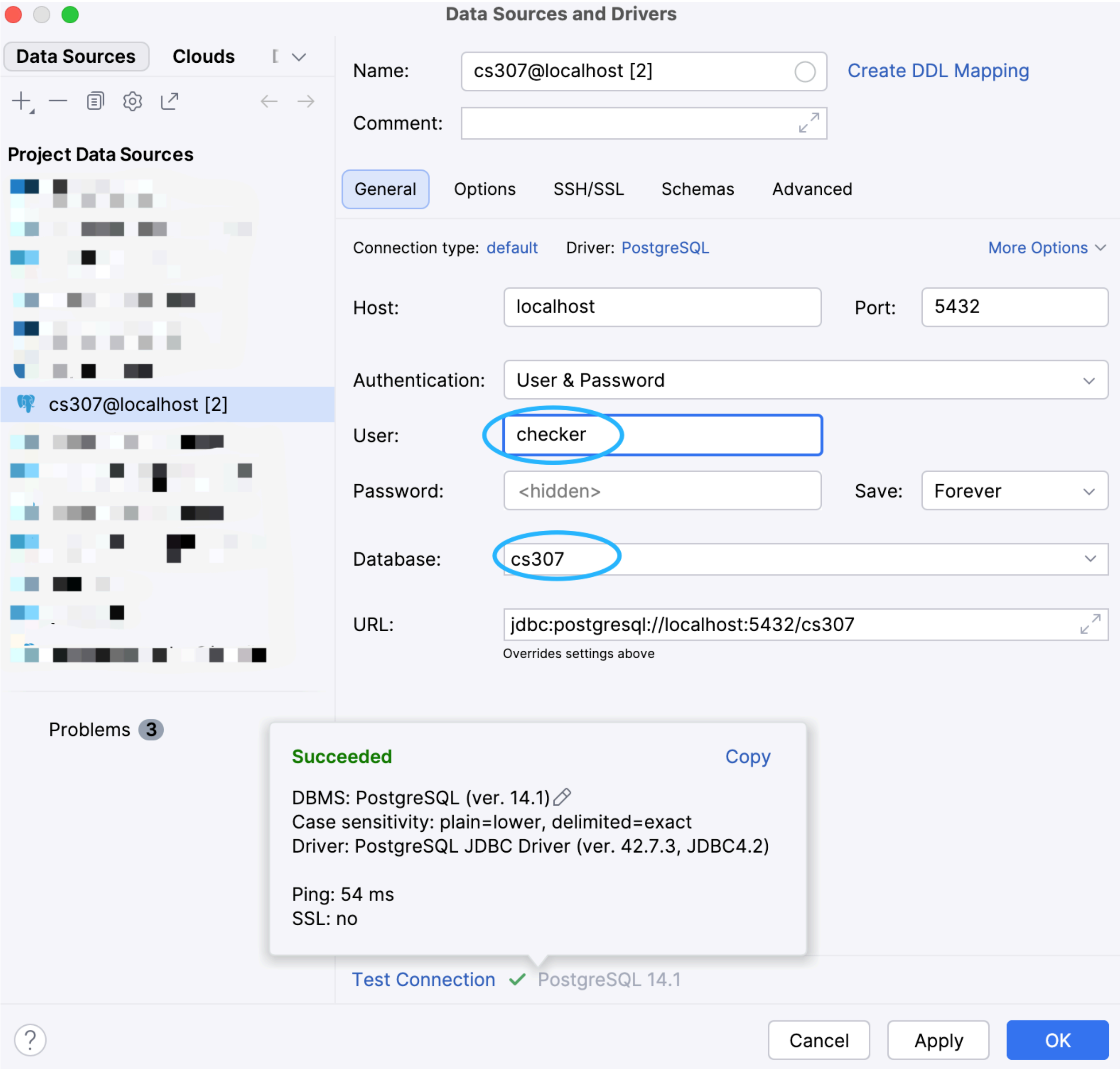


   - Step2

   - Step3



9. **You'd better create another datasource with a new connection to another database `cs307`**



10. **Try to create a table in public schema**

```
create table lab(
    id serial primary key,
    address varchar(20) not null,
    time varchar(20) not null,
    capacity int,
    teacher varchar(20),
    unique (address,time)
);
```

11. **Insert data into lab table**

```
insert into lab (address, time, capacity, teacher) values ('508','2-
78',36,'yueming');
insert into lab (address, time, capacity, teacher) values ('508','2-
56',36,'yueming');
insert into lab (address, time, capacity, teacher) values ('508','3-
56',36,'yueming');
insert into lab (address, time, capacity, teacher) values ('510','4-
34',36,'yueming');
```

Then select to check them

```
select * from lab;
```

12. **Update data**

```
update lab set address='510' where time = '3-34';
```

Then select to check it

```
select * from lab where time = '3-34';
```

13. **Delete data**

```
delete from lab where time = '2-56';
```

Then select to check it

```
select * from lab;
```

Part 3. Exercise

1. Install PostgreSQL in your own computer (any system can be accepted).
2. Create a database named cs307 with a owner named checker.
3. Use your DataGrip to connect database cs307.
4. Access PostgreSQL database by network, i.e. to connect to database cs307 from another IP address instead of 127.0.0.1 , as long as we know your IP address and username (checker). Now, you can search for any solutions to accomplish this task, and I think you, to be a student of CSE, can accomplish this task.

   Hints: Find `pg_hba.conf` `postgresql.conf` file in your psql installation catalog.

   Installation catalog for Linux: `/etc/postgresql/12/main`

   Installation catalog for MacOS: `/usr/local/var/postgres`

   Installation catalog for windows: `C:\Program Files\PostgreSQL\13\data`